

King Saud University
College of Computer and Information Sciences
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Project management IS324

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INTRODUCTION

KSUpay is a payment system designed for King Saud University to manage all financial transactions within in the campus. The system allows the user to create a new account or log in if they already have one, receive monthly stipends, transfer money to other students or any KSU entities. It also allows administrators to manage everything going on, such as viewing the total balance of all KSU entities wallets, deposit monthly stipends to all student wallets, back up all information from the central database to a CSV file.

The system consists of 4 windows(i.e. sign up, log in, student wallet, and admin).

Sign up window: here the user can create an account by entering their information. The system will validate the input, check whether the user already exists and deal with any exceptions.

Log in window: here the user can log into their account if they have one. The system will validate the input, check whether the user has an account and deal with any exceptions.

Student wallet window: here the student can view their wallet number and current balance and transfer money to any other KSU entity. The system will validate the input, check if the balance is enough and do all the calculations.

Admin window: here the admin can get information about the balance of all KSU entities, create a new KSU entity wallet, deposit monthly stipends to all students wallets, reset the balance of all KSU entitie's wallets to zero and get a back up of all data in the central DB.

Challenges

Team-Level Challenges:

A primary challenge we encountered was structuring the project effectively. Initially, we created separate classes for each component (such as Student and Wallet), which led to conflicts when changes in one part affected others. We subsequently reorganized the structure and established a more cohesive design. We also faced syntax and logic errors, particularly during the integration phase. Since tasks were divided among team members, progress often depended on changes from others, making clear communication essential. The most significant challenge involved using Git. As beginners, we frequently encountered merge conflicts, and resolving them without losing work required practice and coordination.

Personal Challenges:

The login window, although simple in concept, presented a challenge in connecting to the database and correctly determining whether the user is an admin or a student, then directing them to the appropriate interface.

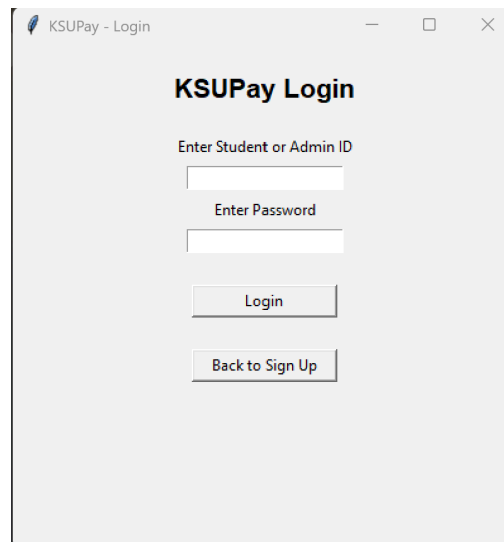
For the sign-up window, the main difficulty was handling database validation - ensuring the correct action is taken if a user already exists. Additionally, deciding on the overall project structure (whether to use an object-oriented approach or integrate logic directly within the GUI) was challenging.

In the Admin window, the most difficult method I implemented was `*CreateWallet*`, due to its logical complexity. While it initially seemed straightforward - taking input and inserting it into the database - handling edge cases made it more complicated. The database was designed with `'owner_id'` referencing students only, but KSU entities (such as the cafeteria) also required wallets. Since these entities do not have numeric IDs, I used their names instead. I then added logic to distinguish between student IDs and entity names, handle empty inputs, prevent duplicate wallets, and resolve SQL-related issues.

For the Student Wallet window, the main challenge was implementing the payment logic correctly. While the concept seemed simple subtracting from one wallet and adding to another the difficulty was in handling all the edge cases in the right order checking if the target wallet exists, verifying the student has sufficient balance, and ensuring both balances are updated correctly in the database so no money is lost or duplicated during the transfer.

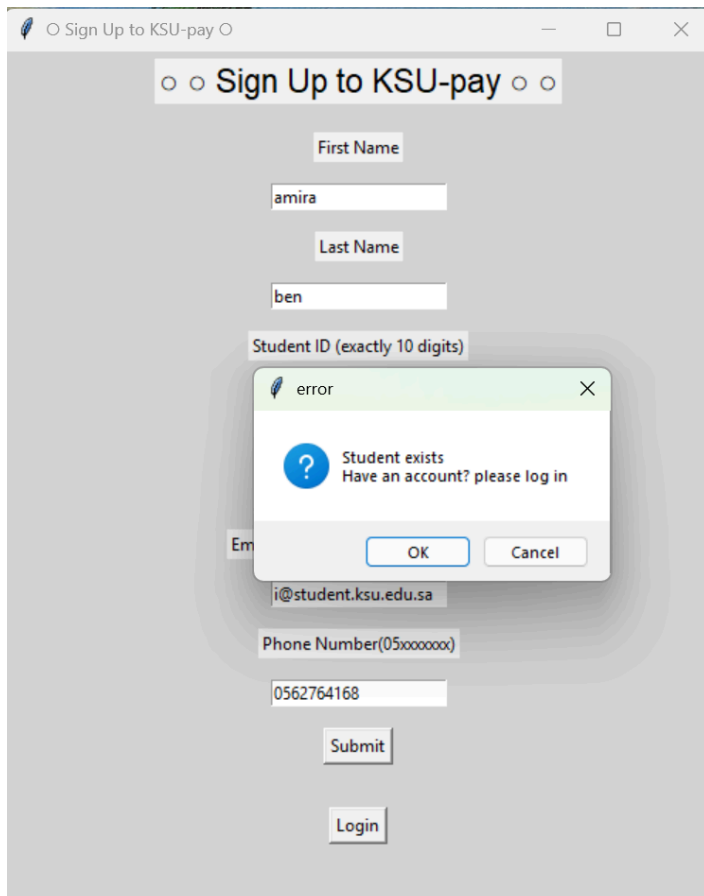
GUI SCREENSHOTS

login window:

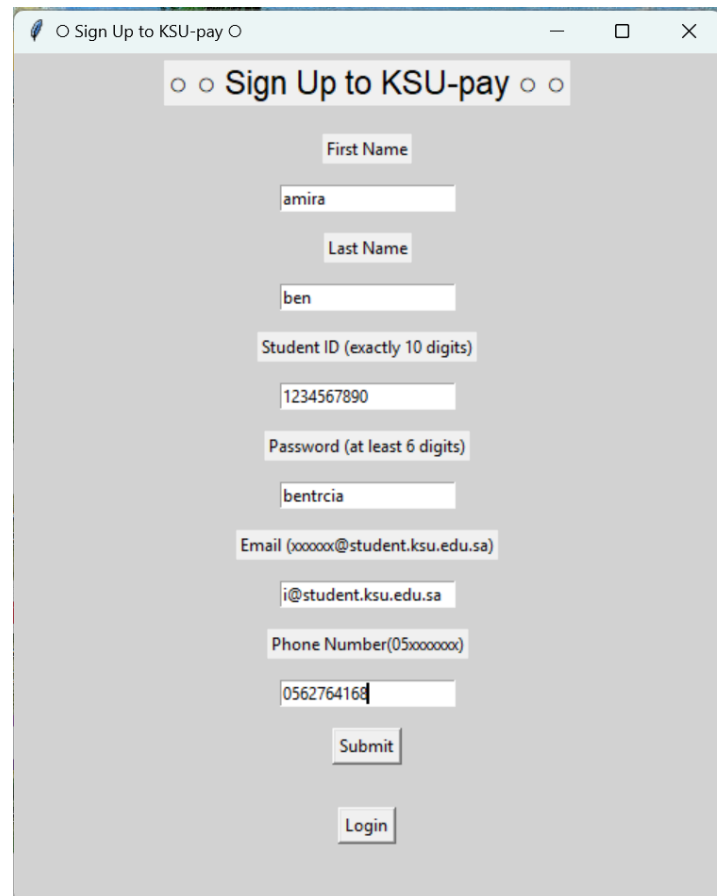


A screenshot of the KSUPay Login window. The window has a title bar with a feather icon and the text "KSUPay - Login". The main content area has a light gray background. At the top, it says "KSUPay Login" in bold. Below that, it says "Enter Student or Admin ID" above a text input field. Then, it says "Enter Password" above another text input field. At the bottom, there are two buttons: "Login" and "Back to Sign Up".

sign up window:



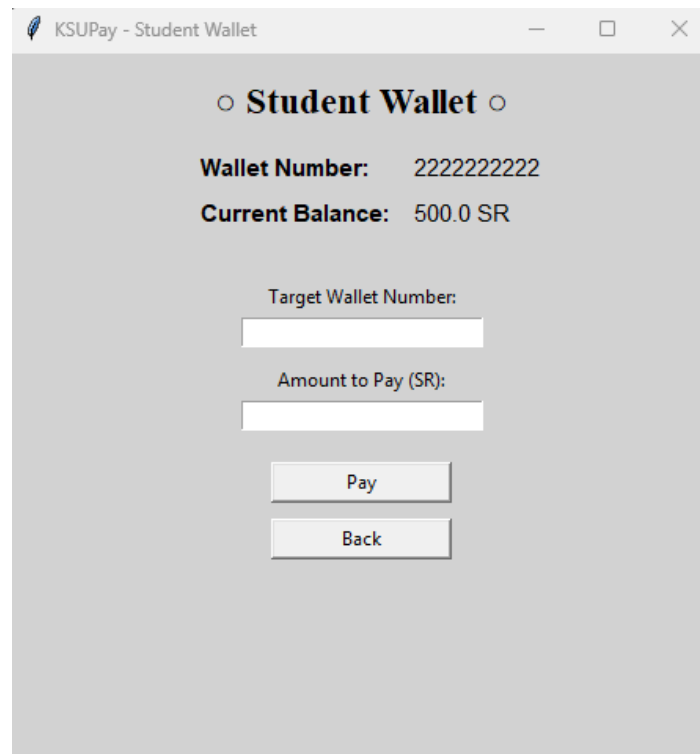
A screenshot of the "Sign Up to KSU-pay" window. The window has a title bar with a feather icon and the text "Sign Up to KSU-pay". The main content area has a light gray background. At the top, it says "Sign Up to KSU-pay" in bold. Below that, there are several text input fields: "First Name" (containing "amira"), "Last Name" (containing "ben"), "Student ID (exactly 10 digits)" (containing "1234567890"), "Password (at least 6 digits)" (containing "bentrcia"), "Email (xxxxxx@student.ksu.edu.sa)" (containing "i@student.ksu.edu.sa"), and "Phone Number(05xxxxxxx)" (containing "0562764168"). At the bottom, there are two buttons: "Submit" and "Login". An error dialog box is overlaid on the form, with a green header and a question mark icon. The text in the dialog says "Student exists. Have an account? please log in". There are "OK" and "Cancel" buttons at the bottom of the dialog.



A screenshot of the "Sign Up to KSU-pay" window, showing the same form as the previous screenshot, but without the error dialog box. The form fields are: "First Name" (amira), "Last Name" (ben), "Student ID (exactly 10 digits)" (1234567890), "Password (at least 6 digits)" (bentrcia), "Email (xxxxxx@student.ksu.edu.sa)" (i@student.ksu.edu.sa), and "Phone Number(05xxxxxxx)" (0562764168). The buttons "Submit" and "Login" are at the bottom.

GUI SCREENSHOTS

Student wallet window:



The screenshot shows a window titled "KSUPay - Student Wallet". The interface has a light gray background. At the top, there is a header "Student Wallet" flanked by two small circles. Below this, the "Wallet Number:" is displayed as "2222222222" and the "Current Balance:" is "500.0 SR". There are two input fields: "Target Wallet Number:" and "Amount to Pay (SR):". Below these fields are two buttons: "Pay" and "Back".

KSUPay - Student Wallet

○ **Student Wallet** ○

Wallet Number: 2222222222

Current Balance: 500.0 SR

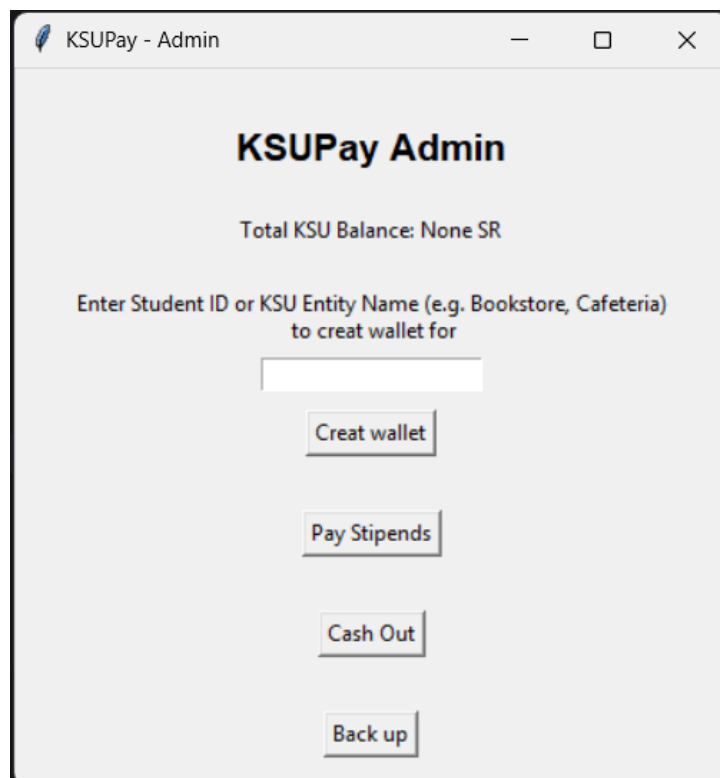
Target Wallet Number:

Amount to Pay (SR):

Pay

Back

Admin window:



The screenshot shows a window titled "KSUPay - Admin". The interface has a light gray background. At the top, there is a header "KSUPay Admin". Below this, the "Total KSU Balance:" is displayed as "None SR". There is a text prompt "Enter Student ID or KSU Entity Name (e.g. Bookstore, Cafeteria) to creat wallet for" followed by an input field. Below the input field are four buttons: "Creat wallet", "Pay Stipends", "Cash Out", and "Back up".

KSUPay - Admin

KSUPay Admin

Total KSU Balance: None SR

Enter Student ID or KSU Entity Name (e.g. Bookstore, Cafeteria)
to creat wallet for

Creat wallet

Pay Stipends

Cash Out

Back up

FLOW CHART DIGRAM

